

Used-car appraisal

Preliminary insight development

COMM5000 · MILESTONE 1 · TERM 3, 2026

Original listings	Primary cohort	Main report words
1,005,000	998,849	1,000

1. Introduction

The client operates an Indian online used-car marketplace where faster, defensible appraisals could improve cash offers and listing-price checks [1]. Overvaluation threatens purchasing margins; undervaluation risks losing sellers. This milestone examines vehicle age, usage and brand segment to prioritise subsequent analysis. It provides preliminary evidence and a development plan rather than a completed pricing model. Only the fixed randomised XLSM is analysed, using descriptive, probability and sampling concepts introduced in Weeks 1–3 [2–4]. Preparation decisions, numerical summaries and graphical relationships inform the recommendations.

2. Data Summaries and Descriptive Statistics

2.1 Data definition and preparation

The dataset [6] contains 1,005,000 listings and 20 variables. Car_Price, recorded in Indian rupees, is the outcome; Registration_Age and Kms_Driven measure supplied age in years and usage in kilometres. Table 2 distinguishes nominal labels, ownership categories, numerical measurements and binary flags. In every observation, Year plus Registration_Age equals 2025. The supplied age is therefore retained as a snapshot measure, rather than recalculated using today's date; including both age and Year later would duplicate information.

Trimming Brand corrects 10,040 labels and recovers 2,334 luxury records that an uncleaned lookup would misclassify. Luxury comprises Audi, BMW and Mercedes; the other ten identified brands constitute non-luxury. Explicit spelling, case and whitespace rules standardise 20,123 Fuel_Type variants. Unknown categories remain visible rather than being guessed. The original workbook contains neither blank cells nor exact duplicates: unique original prices preclude identical complete records, including after text cleaning [5]. Forty zero readings across three mechanical variables are flagged for verification, without imputing unused predictors.

Fuel type and transmission are Unknown for 7.83% and 8.00% of full-file records respectively; dropping those rows would discard usable price, age and usage observations. These unresolved categories will be reported separately when studying additional predictors.

The primary cohort screens 6,151 listings exceeding 1,000,000 kilometres, leaving 998,849 observations. This threshold follows the guide's example of implausible usage but is an appraisal-screening decision, not proof of error. All records remain in the full-sample comparison. High prices are retained: an IQR flag alone cannot establish an incorrect valuation. Table 1 records decisions and denominators.

2.2 Distribution and typical values

The primary mean price is INR 1,015,954, compared with a median of INR 673,944. The substantial gap and Figure 3's right tail indicate positive skew. The price IQR is INR 877,727 and sample SD is INR 1,141,023; dispersion is commercially material. The median is preferable for describing a typical listing because extreme prices pull the mean upward [2]. However, it is not a vehicle-specific appraisal, and the mean remains informative for portfolio averages.

Average age is 13.00 years and median usage is 152,632 kilometres. Table 3 includes means, medians, modes, sample SDs, quartiles and ranges for the full file and screened groups. At original XLSM precision, price and usage have no mode: all values are unique.

Analysis, conclusions and next steps

2.3 Relationships with age and usage

Figures 1–2 use the same 5,000 screened listings selected with Excel RAND(), fixed random values, sorting and retained original row numbers. Sampling makes dense plots legible; correlations use unrounded prices and every eligible pair. The logarithmic price axes display the wide range without discarding expensive listings. The age–price correlation is -0.0012 overall and negligible within both segments. Age-group means in Figure 4 show no systematic depreciation pattern in this randomised version.

Usage has a weak negative linear association with price: $r = -0.0509$ overall, -0.0521 for luxury and -0.0505 for non-luxury. Mean prices decline from INR 1,138,056 to INR 906,258 across the first and last usage deciles, alongside substantial scatter. Hence usage merits further investigation but cannot support precise offers alone. Restoring high-usage records changes the overall correlation to -0.0499 ; the interpretation survives this sensitivity check. Pearson correlation describes linear association, not causation or all possible nonlinear structure [2].

An additional diagnostic removes prices above the upper IQR fence of INR 2,560,035. Usage correlation remains weak at -0.0516 , so the principal conclusion is not driven solely by unusually expensive listings; this diagnostic is not the adopted sample.

2.4 Brand-segment comparison

The screened cohort contains 230,761 luxury and 768,088 non-luxury listings. Their mean prices are INR 1,014,112 and INR 1,016,508 respectively: a luxury difference of -0.236% . Medians differ by only INR -435 , or -0.065% . Figure 3 shows substantial distributional overlap, and both groups have comparable ages and usage. These summaries provide no commercially substantial unconditional luxury premium in this version. They neither establish population equality nor rule out differences conditional on other characteristics. Empirical segment shares describe these listings, not India's market composition [3].

3. Conclusion and Development Plan

The useful preliminary findings are strong price skew, considerable dispersion, negligible age association and a weak usage association. The evidence supports cautious investigation, not automatic depreciation deductions or a fixed luxury surcharge. Listing prices are not verified transaction prices; city coverage, Unknown labels and unresolved measurements limit generalisation. Randomisation may weaken real-world relationships. A million observations can improve precision without removing selection bias: the Central Limit Theorem concerns sample means, not a normal distribution of individual prices or representative sampling [4].

For Milestone 2, define $\Delta = \mu_{\text{luxury}} - \mu_{\text{non-luxury}}$ and examine $H_0: \Delta = 0$ against the two-sided alternative $H_1: \Delta \neq 0$. Report the estimated mean difference in INR, its percentage relative to non-luxury prices, and a confidence interval using subsequently taught methods. Interpret uncertainty and practical relevance together.

Investigate age and usage using within-segment summaries, age groups and usage deciles, checking curvature and whether aggregate patterns persist within brands. State denominators and revisit the mileage screen. For the final report, assess the other supplied variables, including model, fuel type, transmission, engine characteristics, ownership history, city and service history. Compare a pooled model with brand-specific models using the same reserved observations. Report mean absolute error and root mean squared error in INR, overall, by brand and for expensive listings; compare against a simple benchmark and avoid information leakage. Verify flagged measurements and transaction-price limitations before operational recommendations. M1 proposes these analyses without performing hypothesis tests or fitting predictive models. Uncertain appraisals should remain subject to human review.

Evidence A · Preparation and audit

Table 1a. Original-XLSM preparation audit

Audits use all 1,005,000 source records at stored precision. Original DATA and Dataset!B6 are preserved. Text corrections do not change numerical values. No blanket complete-case deletion or imputation is applied.

Audit item	Observed count	Treatment / rationale
Source observations	1,005,000	Original DATA rows; source SHA256 recorded.
Blank cells	0	Across all 20 variables; Unknown labels counted separately.
Exact raw duplicate records	0	Unique original prices preclude identical complete records.
Duplicates after text cleaning	0	Numeric prices remain unchanged and unique.
Brand labels corrected	10,040	TRIM removes spaces before segment construction.
Luxury classifications recovered	2,334	Uncleaned lookup would misclassify these records.
Fuel label variants standardised	20,123	Six explicit case/spelling/space rules; Unknown retained.
Kms_Driven > 1,000,000	6,151 (0.612%)	Screen in primary results; keep in full-sample sensitivity.
Primary cohort	998,849	No other observation exclusions.
Zero fuel-efficiency / engine / power	5 / 12 / 23	Flag unresolved auxiliary readings; do not impute.
Primary high-price IQR flags	70,457	Above INR 2,560,035; retain in primary results.

Table 1b. Unknown categories retained in the full file

Unresolved category	Unknown count	Full-file share (%)
Fuel_Type	78,733	7.83
Transmission	80,385	8.00
Color	80,419	8.00
City	80,430	8.00

Table 1c. Cohort reconciliation after Brand normalisation

Cohort	All	Luxury	Non-luxury
Full	1,005,000	232,172	772,828
High-usage screen	6,151	1,411	4,740
Primary	998,849	230,761	768,088

Evidence B · Variables and units

Table 2. Definitions, measurement types and analytical roles

Variable	Type / unit	Meaning and treatment
Brand	Nominal	Vehicle manufacturer; cleaned before segment construction.
Model	Nominal	Model name; candidate predictor for later analysis.
Year	Integer / year	Manufacturing year, 2000–2024. Age + Year = 2025 in every row.
Mileage_kmpl	Continuous / km per litre	Fuel efficiency; five zero values flagged.
Engine_CC	Continuous / cm ³	Engine displacement; twelve zeros require contextual checking.
Horsepower	Continuous / reported horsepower	Engine power output; twenty-three zero readings flagged.
Fuel_Type	Nominal	Petrol, Diesel, CNG, Electric, Hybrid, Unknown after correction.
Transmission	Nominal	Manual, Automatic, Unknown.
Owner_Type	Ordinal category	First, Second, Third, Fourth+ owner category.
Color	Nominal	Vehicle colour; Unknown retained.
City	Nominal	Registration city; eight named cities and Unknown.
Kms_Driven	Continuous / km	Total distance travelled; key usage variable.
Insurance_Valid	Binary / 0 or 1	Recorded insurance-status flag; 1 indicates valid insurance.
Service_History	Binary / 0 or 1	Recorded service-history availability flag.
Accidents	Integer / count	Number of reported accidents.
Tax_Paid	Binary / 0 or 1	Recorded road-tax-payment flag.
Number_of_Doors	Integer / count	Vehicle door count.
Seats	Integer / count	Seating capacity.
Registration_Age	Integer / supplied years	Key age measure, 1–25; retain the supplied snapshot definition.
Car_Price	Continuous / INR	Reported price in rupees; outcome, not verified transaction proceeds.

Source: the original Dataset worksheet [6] and course dataset introduction [5, PDF p. 8]. INR denotes Indian rupees. Supplied age is retained: Year + Registration_Age = 2025 holds for every observation.

Evidence C · Numerical summaries

Table 3. Full file and primary cohort, overall and by segment

F = full supplied file; S = primary cohort (Kms_Driven ≤ 1,000,000). Luxury = Audi, BMW, Mercedes after Brand TRIM. Prices and usage are shown in thousands; sample SD uses n – 1. All statistics use original unrounded values; only presentation is rounded.

Table 3a. Car_Price (INR thousands)

Cohort / segment	n	Mean	Median	SD	Min	Max	Mode
F: All	1,005,000	1,014.8	673.1	1,140.1	10.4	46,802.5	No mode†
F: Luxury	232,172	1,013.0	672.8	1,129.1	11.5	46,802.5	No mode†
F: Non-luxury	772,828	1,015.3	673.2	1,143.4	10.4	38,036.4	No mode†
S: All	998,849	1,016.0	673.9	1,141.0	10.4	46,802.5	No mode†
S: Luxury	230,761	1,014.1	673.6	1,130.3	11.5	46,802.5	No mode†
S: Non-luxury	768,088	1,016.5	674.0	1,144.2	10.4	38,036.4	No mode†

Table 3b. Registration_Age (years)

Cohort / segment	n	Mean	Median	SD	Min	Max	Mode
F: All	1,005,000	13.0	13.0	7.2	1.0	25.0	6
F: Luxury	232,172	13.0	13.0	7.2	1.0	25.0	18
F: Non-luxury	772,828	13.0	13.0	7.2	1.0	25.0	19
S: All	998,849	13.0	13.0	7.2	1.0	25.0	6
S: Luxury	230,761	13.0	13.0	7.2	1.0	25.0	18
S: Non-luxury	768,088	13.0	13.0	7.2	1.0	25.0	19

Table 3c. Kms_Driven (thousand km)

Cohort / segment	n	Mean	Median	SD	Min	Max	Mode
F: All	1,005,000	203.0	153.5	175.9	4.5	6,259.5	No mode†
F: Luxury	232,172	202.9	153.2	175.7	4.5	4,129.8	No mode†
F: Non-luxury	772,828	203.1	153.6	175.9	4.7	6,259.5	No mode†
S: All	998,849	196.2	152.6	150.7	4.5	999.9	No mode†
S: Luxury	230,761	196.1	152.4	150.9	4.5	999.5	No mode†
S: Non-luxury	768,088	196.2	152.7	150.6	4.7	999.9	No mode†

† No mode — all values unique. MODE.MULT returns no repeated value for Car_Price and Kms_Driven in every displayed cohort at original XLSM precision. Age modes are actual year values, not numbers of tied modes.

Table 3d. Primary-cohort quartiles and interquartile ranges

Variable / unit	All: Q1 / Q3 / IQR	Luxury: Q1 / Q3 / IQR	Non-luxury: Q1 / Q3 / IQR
Price / INR thousands	365.7 / 1,243.4 / 877.7	365.0 / 1,246.6 / 881.5	365.9 / 1,242.6 / 876.6
Age / years	7.0 / 19.0 / 12.0	7.0 / 19.0 / 12.0	7.0 / 19.0 / 12.0
Usage / thousand km	92.4 / 251.7 / 159.3	92.4 / 251.4 / 158.9	92.4 / 251.8 / 159.4

Quartiles use Excel QUARTILE.INC. IQR = Q3 – Q1. The complete analysis workbook retains the source formulas and fixed native results used here, alongside the source data and sampling evidence.

Evidence D · Relationships

Table 4. Pearson correlations from all eligible pairs

Cohort / segment	Eligible pairs	r: age-price	r: usage-price
F: All	1,005,000	-0.0012	-0.0499
F: Luxury	232,172	-0.0009	-0.0508
F: Non-luxury	772,828	-0.0013	-0.0497
S: All	998,849	-0.0012	-0.0509
S: Luxury	230,761	-0.0008	-0.0521
S: Non-luxury	768,088	-0.0013	-0.0505
Extra price screen*	928,392	-0.0005	-0.0516

* Diagnostic only: additionally omit 70,457 prices above INR 2,560,035 ($Q3 + 1.5 \times IQR$). Primary estimates retain those prices. No hypothesis tests are performed.

Figures 1–2. Price versus age and usage

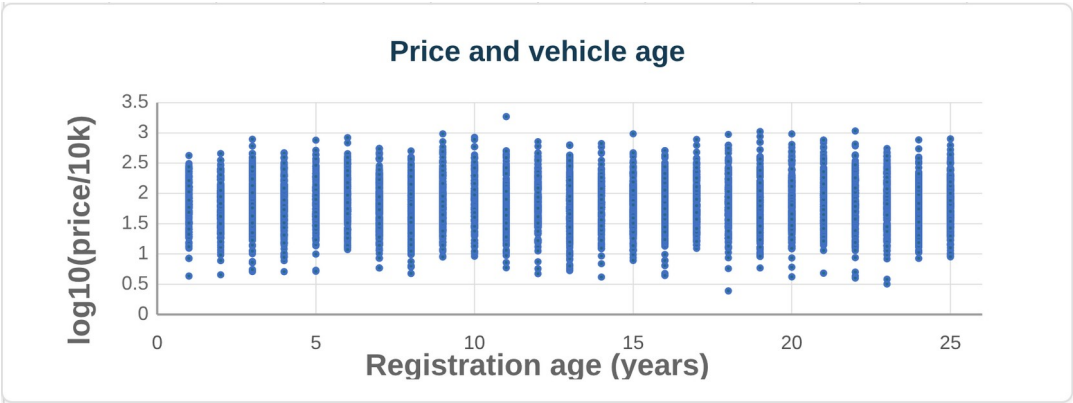


Figure 1. Original age versus price; fixed random sample, n = 5,000.

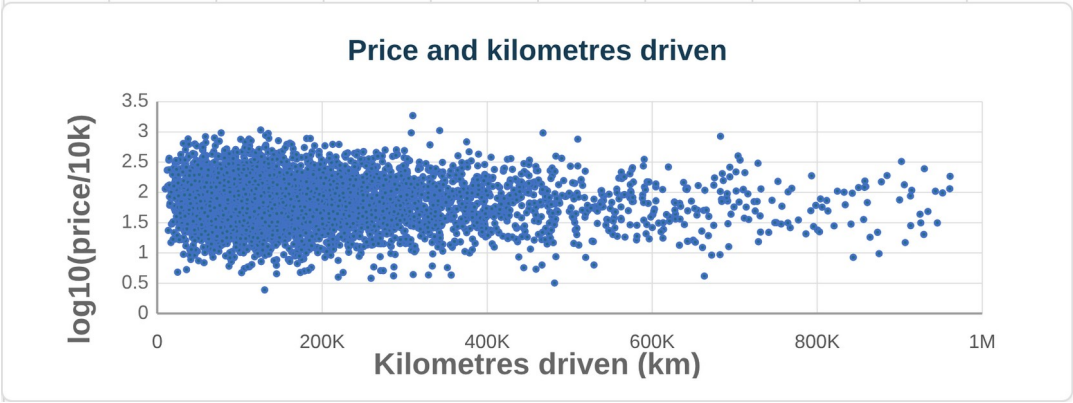


Figure 2. Original kilometres driven versus price; the identical sample.

Vertical coordinate = $\log_{10}(\text{price} / \text{INR } 10,000)$: 1 = INR 100,000; 2 = INR 1 million; 3 = INR 10 million. Every sampled price is retained.

Sample: 1,194 luxury and 3,806 non-luxury listings, all unique and eligible. Original DATA row IDs and fixed RAND values are retained. Correlations above use full eligible cohorts and untransformed prices.

Evidence E · Distributions and grouped patterns

Figure 3. Primary-cohort frequencies within each segment. Bins are INR 250,000 wide; x labels give lower bounds. The final $\geq$ INR 5 million bin includes the full upper tail.

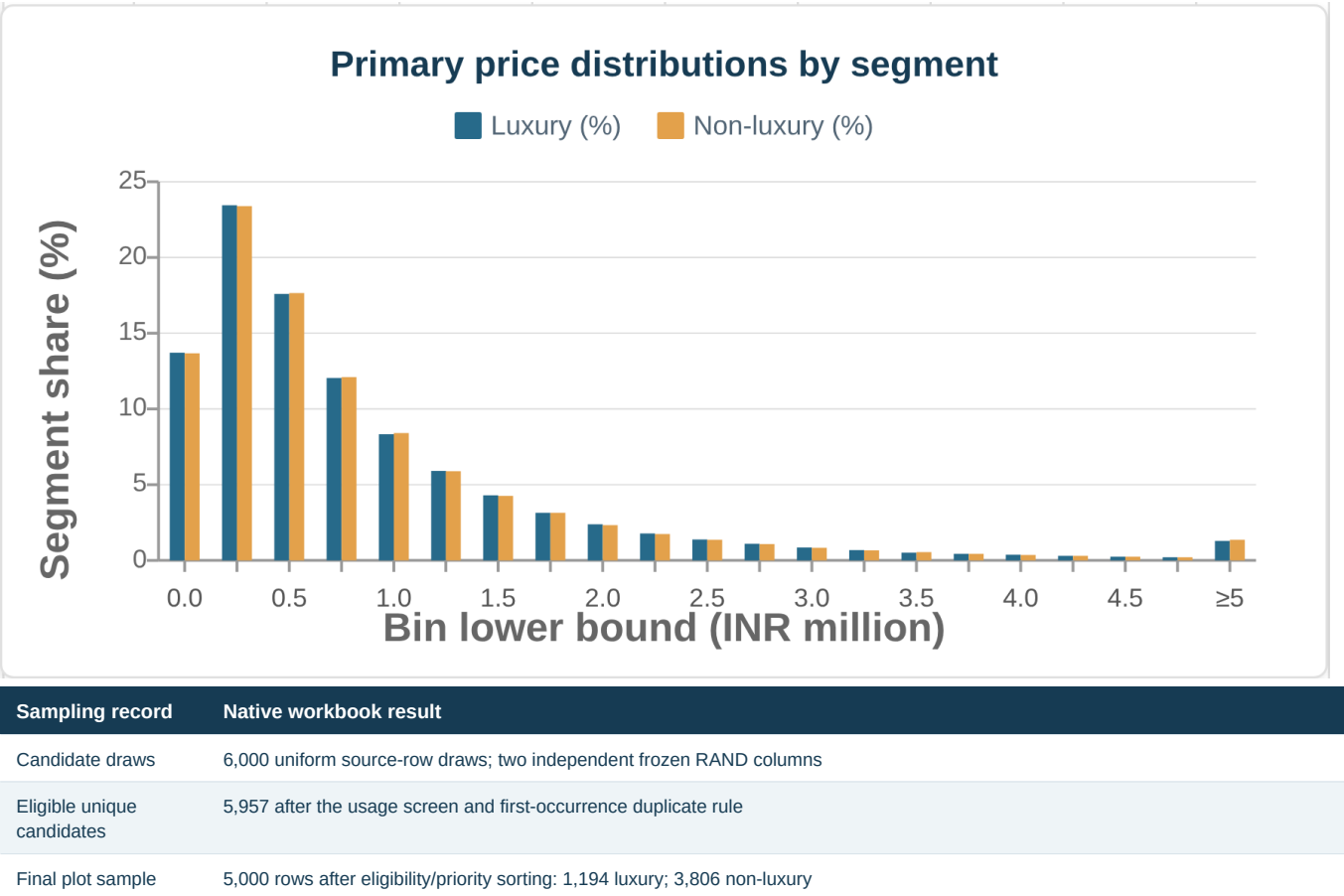
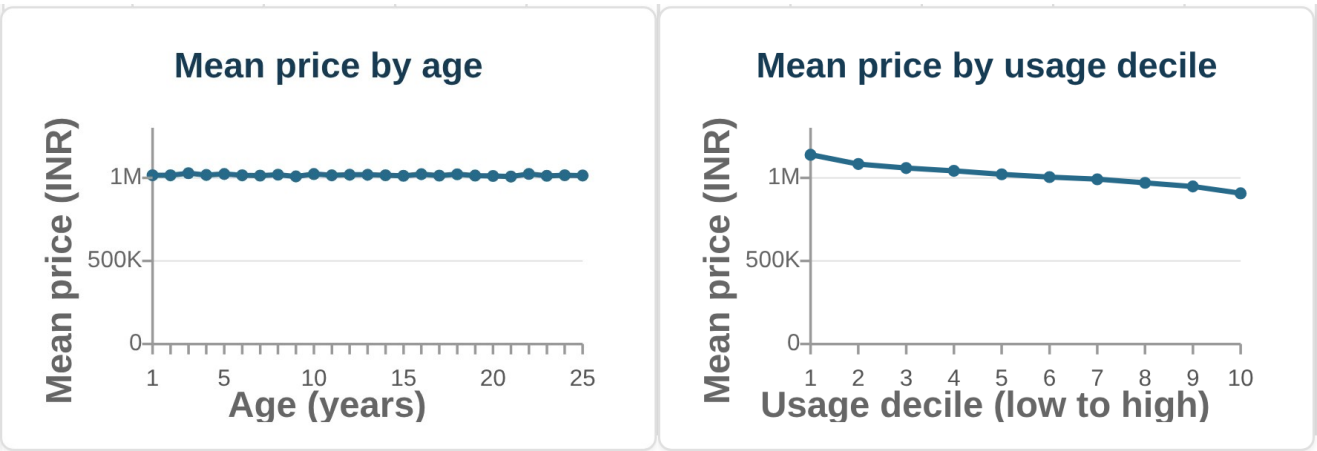


Figure 4. Primary-cohort group means

The left panel uses each supplied age year; the right uses usage deciles. Every eligible observation contributes to its group mean. Mean curves describe groups and do not show individual prediction accuracy.



References and calculation record

[1] UNSW Business School (2026). COMM5000 T3 2026 Assessment Guide Milestone 1 – Used Cars. PDF pp. 6, 8–11: business scenario, M1 tasks, structure, submission requirements and marking rubric.

[2] UNSW Business School (2026). COMM5000–Week 1–T326. PDF pp. 27–55: data quality, sampling, graphical and numerical summaries, dispersion and Pearson correlation.

[3] UNSW Business School (n.d.). COMM5000 W2 Lecture. PDF pp. 10, 17–22, 32–34: empirical and conditional probability; probability distributions.

[4] UNSW Business School (n.d.). COMM5000 W3 Lecture. PDF pp. 4–11: descriptive/inferential statistics, estimators, sampling distributions and the Central Limit Theorem. Historical administrative dates are not used.

[5] UNSW Business School (2026). COMM5000_Course-Dataset-M1-intro-T326. PDF pp. 2–8: project scope, workbook use, dataset-description screenshot and randomisation guidance.

[6] COMM5000 course dataset (2026). COMM5000-Used_Car_Price_Prediction.xlsm. Original randomised workbook, 1,005,000 observations and 20 variables, read at stored precision. Dataset!B6 records randomisation on 2026-10-07 at 06:20:30 (timezone unspecified). Original publisher: Stable Space (2026), Used Car Price Prediction Dataset, <https://www.kaggle.com/datasets/sharmajicoder/used-car-price-prediction-dataset> (address recorded in the supplied workbook; public data not substituted).

Course-material access: <https://moodle.telt.unsw.edu.au/> (login required). Exact filenames and physical PDF page locations above identify the supplied materials. Individual Moodle resource URLs were not provided.

Source and calculation record

Item	Record
Numerical source	Original randomised XLSM; DATA and Dataset retained at stored precision. No CSV used in this analysis.
Randomisation record	Dataset!B6: applied on 2026-10-07 06:20:30; the original status text is preserved.
Price / usage modes	No mode — all values unique; independently calculated on original-precision values in all six cohorts.
Computation	Native Excel formulas automated with OfficeCLI 1.0.155. Source formulas and fixed calculation snapshots are retained.
Charts	Native Excel chart objects, frozen RAND selection, source row IDs and chart ranges retained in the complete workbook.
Sensitivity	Full file, primary usage screen and additional price-IQR diagnostic reported separately; high prices retained in primary results.
Main-report word count	1,000 words including section headings; title, metadata, tables, figures/captions and references excluded.